

Fujie Tang (汤富杰), Ph.D.

CONTACT INFORMATION	Room B308, Zengchengkui Building, Siming Campus, Xiamen University, Siming District, Xiamen, Fujian 361005, China Research Group: https://fujiepkku.github.io University Profile: https://chem.xmu.edu.cn/info/1421/126815.htm	E-mail: tangfujie@xmu.edu.cn ORCID: 0000-0001-9761-5359
ACADEMIC POSITIONS	Professor, PI College of Chemistry and Chemical Engineering, Institute of Artificial Intelligence, Xiamen University, Xiamen, Fujian, China Research Professor Laboratory of AI for Electrochemistry (AI4EC Lab), Tan Kah Kee Innovation Laboratory (嘉庚创新实验室, IKEEM), Xiamen, Fujian, China Professor Pen-Tung Sah Institute of Micro-Nano Science and Technology, Institute of Artificial Intelligence, Xiamen University, Xiamen, Fujian, China Associate Professor Pen-Tung Sah Institute of Micro-Nano Science and Technology, Institute of Artificial Intelligence, Xiamen University, Xiamen, Fujian, China Associate Research Professor Laboratory of AI for Electrochemistry (AI4EC Lab) Tan Kah Kee Innovation Laboratory (嘉庚创新实验室, IKEEM), Xiamen, Fujian, China	2026.07-Present 2025.12-Present 2025.12-2026.07 2024.01-2025.12 2024.01-2025.12
POSTDOCTORAL AND VISITING EXPERIENCE	CSI Postdoctoral Scholar Computational Chemical Science Center, Princeton University, Princeton, NJ, USA Supervisor: Prof. Xifan Wu & Prof. Roberto Car Visiting Scholar Department of Physics, University of California at Berkeley, California, CA, USA Supervisor: Prof. Steven G. Louie Postdoctoral Fellow	2019.10-2023.10 2019.06-2019.07 2018.10-2019.09

Department of Physics,
Temple University, Philadelphia, PA, USA
Supervisor: Prof. Xifan Wu

Visiting Ph.D. Student

2019.06-2019.07

Department of Molecular Spectroscopy,
Max Planck Institute for Polymer Research, Mainz, Germany
Supervisor: Dr. Yuki Nagata & Prof. Mischa Bonn

EDUCATION

Peking University, Beijing, China

Ph.D., Condensed Matter Physics

2013.09-2018.06

- Thesis Topic: *Structures and Dynamics of Interfacial Water*
- Supervisor: Prof. Limei Xu

Peking University, Beijing, China

B.S., Major in Physics

2009.09-2013.07

- Advisor: Prof. Tongjun Yu

RESEARCH
INTERESTS

1. Computational Spectroscopy: Method Development and Mechanistic Interpretation

- Vibrational spectroscopy: Development of computational methods for vibrational spectroscopy of complex interfaces and solution systems, with a particular focus on extracting molecular-level structural and interaction information from infrared, Raman, and vibrational sum-frequency generation spectra
- Excited-state spectroscopy: Development of theoretical spectroscopic methods for electronic and core-level excitations, with a particular focus on X-ray absorption and ultraviolet/visible absorption spectroscopy, to elucidate electronic structure, excitonic effects, and charge-transfer mechanisms

2. Theoretical and Computational Chemistry: Method Development and Applications to Complex Chemical Systems

- Electronic-structure methods: Development of ground-state electronic-structure methods integrating concepts from density functional theory and quantum chemistry, together with excited-state approaches such as GW-BSE, to investigate chemical bonding, electronic structure, interfacial charge transfer, and excitonic effects
- Atomic and molecular simulations: Development and application of first-principles molecular dynamics and accelerated simulation methods to investigate structural evolution, molecular interactions, dynamical processes, and nuclear quantum effects in complex chemical environments

3. **Structure, Dynamics, and Spectroscopy of Complex Chemical Systems**
 - Investigation of the structure and dynamics of complex solid–liquid interfaces, electrochemical interfaces, confined water systems, and low-dimensional material interfaces, with an emphasis on interfacial hydrogen-bond networks, ion solvation, charge transfer, and their spectroscopic signatures
 - Investigation of the structure, dynamics, phase behavior, and spectroscopic properties of water and aqueous systems under extreme conditions and nanoscale confinement
4. **Artificial Intelligence for Theoretical Chemistry, Computational Spectroscopy, and Electrochemistry**
 - Development of physics-informed and interpretable machine-learning methods to accelerate molecular simulations, spectroscopic calculations, and structural characterization of complex chemical systems
 - Construction of “structure–dynamics–spectroscopy” relationship models for complex chemical systems to enable experimental spectral interpretation, mechanistic understanding, and AI-assisted spectroscopic characterization

FUNDING	• Excellent Young Scientists Fund (Overseas)	PI
	National Natural Science Foundation of China	2026.01–2028.12
	• General Program	PI
	National Natural Science Foundation of China	2026.01–2029.12
	• National Key R&D Program —Catalytic Science	co-PI
AWARDS AND HONORS	Ministry of Science and Technology of China	2025.12–2030.11
	• National Key R&D Program —Nano Frontiers	co-PI
	Ministry of Science and Technology of China	2024.12–2029.11
	• Natural Science Foundation Youth Project	PI
	Xiamen Municipal	2024.07–2027.06
AWARDS AND HONORS	• National High-level Talent Program (Youth Project)	2025.11
	• High-level Overseas Returnee of Xiamen	2025.07
	• The “Top Talent Recruitment Program” of Fujian	2024.12
	• Springer Thesis Award of 2018, Springer Nature Singapore	2018.08
	• The Excellent Doctoral Dissertation of Peking University	2018.07
	• Distinguished Graduate of Peking University	2018.07

Highlights: 33 published papers, first/co-first/corresponding/co-corresponding author papers = 21, cited 2130 times, h-index = 18. (Updated: Sept.-06-2026)

Google Scholar: <https://scholar.google.com/citations?user=-Rx1hiIAAAAJ&hl=en>

1. Suyang Zhong[#], Yuhao Zhao[#], Boying Huang, Fanjie Xu, Pengwei Xu, Haoyi Tao, Xi Fang*, Jun Cheng*, **Fujie Tang***. Uni-XAS: Alignment-Driven Bidirectional Multimodal Learning for X-ray Absorption Spectroscopy. *34th ACM International Conference on Multimedia (ACM MM)*. Accepted. (*corresponding author)
2. Mingyang Xia[#], Xianglong Du[#], Canbin Wang, Jinguo Liu, Zhihua Zhou, Ying-Ru Qiu, Yao-Hui Wang, **Fujie Tang***, Sheng Hu*. Ion-disrupted hydrogen-bond networks enable fast water transport under two-dimensional confinement. *Nat. Commun.*, 2026, **17**, 8585. (*corresponding author)
3. Xianglong Du, Jun Cheng*, **Fujie Tang***. Machine-Learning Accelerated Computational Spectroscopy Reveals Vibrational Signature of the Oxidation Level of Graphene in Contact with Water. *J. Phys. Chem. Lett.*, 2026, **17**, 1471-1478. (*corresponding author)
4. Kang Wang, Yingchen Peng, Boying Huang, Chun Zhou, Qianlu Sun, **Fujie Tang***, Weigao Xu, Kezhao Du, Xingzhi Wang, Ye Yang*. Correlation between Exciton Dynamics and Spin Structure in van der Waals Antiferromagnet NiPS₃. *Phys. Rev. B(Letter)*, 2025, **112**, L220413. (*corresponding author)
5. Yongkang Wang[#], **Fujie Tang[#]**, Xiaoqing Yu, Kuo-Yang Chiang, Chun-Chieh Yu, Tatsuhiko Ohto, Yunfei Chen, Yuki Nagata, Mischa Bonn. Interfaces Govern Structure of Angstrom-scale Confined Water Solutions. *Nat. Commun.*, 2025, **16**, 7288. ([#]equal contribution)
6. Xianglong Du, Qi You, Jiezhen Xia, **Fujie Tang***, Jun Cheng*, Zhongqun Tian. Progress in machine learning-accelerated computational spectroscopy. *Sci. Sin. Chim.*, 2025, **55**. (*corresponding author) (Invited Review) (In Chinese)
7. Hui Li, Wenqi Zheng, Xinyi Liu, Jiashi Li, Lianbing Wen, **Fujie Tang** and Weitao Liu. Auto-dissociation of atmospheric water on TiO₂: insights from sum-frequency spectroscopy of Ti-O vibrations. *Phys. Chem. Chem. Phys.*, 2025, **27**, 9991-9997
8. Qi You, Yan Sun, Feng Wang, Jun Cheng*, **Fujie Tang***. Decoding the Competing Effects of Dynamic Solvation Structures on NMR Chemical Shifts of Battery Electrolytes via Machine Learning. *J. Am. Chem. Soc.*, 2025, **147**, 14667 - 14676. (*corresponding author)
9. Fanjie Xu, Wentao Guo, Feng Wang, Lin Yao, Hongshuai Wang, **Fujie Tang***, Zhifeng Gao*, Linfeng Zhang, Weinan E, Zhong-Qun Tian, Jun Cheng*. Towards a Unified Benchmark and Framework for Deep Learning-Based Prediction of Nuclear Magnetic Resonance Chemical Shifts. *Nat. Comput. Sci.*, 2025, **15**, 292-300. (*corresponding author)

10. **Fujie Tang**, Diana Y. Qiu and Xifan Wu. Optical Absorption Spectroscopy Probes Water Wire and Its Ordering in a Hydrogen-bond Network. *Phys. Rev. X*, 2025, **15**, 011048.
Featured news in Viewpoint of *Physics* magazine “Shedding Light on Water Wires” written by Davide Donadio and Giulia Galli.
11. Xianglong Du, Weizhi Shao, Chenglong Bao, Linfeng Zhang, Jun Cheng*, **Fujie Tang***. Revealing the Molecular Structures of α -Al₂O₃ (0001)-water Interface by Machine Learning Based Computational Vibrational Spectroscopy. *J. Chem. Phys.*, 2024, **161**, 124702. (*corresponding author) (**JCP Editor’s Pick**)
12. Yongkang Wang[#], **Fujie Tang[#]**, Xiaoqing Yu, Tatsuhiko Ohto, Yuki Nagata, Mischa Bonn. Heterodyne-Detected Sum-Frequency Generation Vibrational Spectroscopy Reveals Aqueous Molecular Structure at the Suspended Graphene/Water Interface. *Angew. Chem. Int. Ed.*, 2024, **63**, e202319503. ([#]equal contribution)
13. Yu Qiao, Ren Hu, Yu Gu, **Fujie Tang**, Siheng Luo, Haitang Zhang, Jinghua Tian, Jun Cheng, Zhongqun Tian. Exploring New Generation of Characterization Approaches for Energy Electrochemistry –from Operando to Artificial Intelligence. *Sci. Sin. Chim.*, 2024, **54**, 338-352. (In Chinese)
14. **Fujie Tang***, Kefeng Shi and Xifan Wu. Exploring the Impact of Ions on Oxygen K-Edge X-ray Absorption Spectroscopy in NaCl Solution using the GW-Bethe-Salpeter-Equation Approach. *J. Chem. Phys.*, 2023, **159**, 174501. (*corresponding author)
15. Chun-Chieh Yu, Takakazu Seki, Kuo-Yang Chiang, **Fujie Tang**, Shumei Sun, Mischa Bonn and Yuki Nagata. Polarization-Dependent Heterodyne-Detected Sum Frequency Generation Spectroscopy as a Tool to Explore Surface Molecular Orientation and Ångströmscale Depth Profiling. *J. Phys. Chem. B*, 2022, **126**, 6113-6124.
16. **Fujie Tang**, Zhenglu Li, Chunyi Zhang, Steven G. Louie, Roberto Car, Diana Y. Qiu and Xifan Wu. Many-body Effects in the X-ray Absorption Spectra of Liquid Water. *Proc. Natl. Acad. Sci. U.S.A.*, 2022, **119**, e2201258119.
17. Hongwei Wang, **Fujie Tang**, Massimiliano Stengel, Hongjun Xiang, Qi An, Tony Low, and Xifan Wu. Convert Widespread Paraelectric Perovskite to Ferroelectrics. *Phys. Rev. Lett.*, 2022, **128**, 197601.
18. Chunyi Zhang, **Fujie Tang**, Mohan Chen, Linfeng Zhang, Diana Y. Qiu, John P. Perdew, Michael L. Klein, Xifan Wu. Modeling Liquid Water by Climbing up Jacob’s Ladder in Density Functional Theory Facilitated by Using Deep Neural Network Potentials. *J. Phys. Chem. B*, 2021, **125**, 11444-11456.
Selected as a Cover Art Article.

19. **Fujie Tang**, Jianhang Xu, Diana Y. Qiu and Xifan Wu. Nuclear Quantum Effects on the Quasiparticle Properties of the Chloride Anion Aqueous Solution within the GW Approximation. *Phys. Rev. B*, 2021, **104**, 035117.
20. Chunyi Zhang, Linfeng Zhang, Jianhang Xu, **Fujie Tang**, Biswajit Santra, and Xifan Wu, Isotope Effects in X-ray Absorption Spectra of Liquid Water, *Phys. Rev. B*, 2020, **102**, 115155.
21. Hongwei Wang, **Fujie Tang**, Pratikkumar H. Dhuvad, and Xifan Wu, Interface Enhanced Functionalities in Oxide Superlattices under Mechanical and Electric Boundary Conditions, *npj Comput Mater*, 2020, **6**, 52.
22. **Fujie Tang**, Tatsuhiko Ohto, Shumei Sun, Jeremy R. Rouxel, Sho Imoto, Ellen H. G. Backus, Shaul Mukamel, Mischa Bonn, and Yuki Nagata. Molecular Structure and Modeling of Water-Air and Ice-Air Interfaces Monitored by Sum-frequency Generation. *Chem. Rev.*, 2020, **120**, 3633-3667.
23. **Fujie Tang**, Xuanyuan Jiang, Hsin-Yu Ko, Jianhang Xu, Mehmet Topsakal, Guanhua Hao, Alpha T. N'Diaye, Peter A. Dowben, Deyu Lu, Xiaoshan Xu, and Xifan Wu. Probing Ferroelectricity by X-ray Absorption Spectroscopy in Molecular Crystals. *Phys. Rev. Materials*, 2020, **4**, 034401.
24. Tatsuhiko Ohto, Mayank Dodia, Jianhang Xu, Sho Imoto, **Fujie Tang**, Frederik Zysk, Thomas D. Kuhne, Yasuteru Shigeta, Mischa Bonn, Xifan Wu, Yuki Nagata. Accessing the Accuracy of Density Functional Theory through Structure and Dynamics of the Water-Air Interface. *J. Phys. Chem. Lett.*, 2019, **123**, 099602.
25. Ruidan Zhang, Jichao Dong, Ting Luo, **Fujie Tang**, Xingxing Peng, Chuanyao Zhou, Xueming Yang, Limei Xu, Zefeng Ren. Adsorption Structure and Coverage-Dependent Orientation Analysis of Submonolayer Acetonitrile on $\text{TiO}_2(110)$. *J. Phys. Chem. C*, 2019, **123**, 17915-17924.
26. Shumei Sun[#], **Fujie Tang**[#], Sho Imoto, Daniel R. Moberg, Tatsuhiko Ohto, Francesco Paesani, Mischa Bonn, and Yuki Nagata. Orientational Distribution of Free OH Groups of Interfacial Water is Exponential. *Phys. Rev. Lett.*, 2018, **121**, 246101. ([#]equal contribution)
27. Bart Weber, Yuki Nagata, Stephania Ketzetzi, **Fujie Tang**, Wilbert J. Smit, Huib J. Bakker, Ellen H. G. Backus, Mischa Bonn, and Daniel Bonn. Molecular Insight into the Slipperiness of Ice. *J. Phys. Chem. Lett.*, 2018, **9**, 2838.
Highlighted by Nature “Why your feet slip and slide on ice?”
28. **Fujie Tang**, Tatsuhiko Ohto, Taisuke Hasegawa, Wen Jun Xie, Limei Xu, Mischa Bonn, and Yuki Nagata. Definition of Free O-H Groups of Water at the Air-Water Interface. *J. Chem. Theory Comput.*, 2018, **14**, 357.

29. Wilbert J. Smit[#], **Fujie Tang**[#], M. Alejandra Sanchez, Ellen H. G. Backus, Limei Xu, Taisuke Hasegawa, Mischa Bonn, Huib J. Bakker, and Yuki Nagata. Excess Hydrogen Bond at the Ice-Vapor Interface around 200 K. *Phys. Rev. Lett.*, 2017, **119**, 133003. ([#]equal contribution)
30. Wilbert J. Smit, **Fujie Tang**, Yuki Nagata, M. Alejandra Sanchez, Taisuke Hasegawa, Ellen H. G. Backus, Mischa Bonn, and Huib J. Bakker. Observation and Identification of a New OH Stretch Vibrational Band. *J. Phys. Chem. Lett.*, 2017, **8**, 3656.
31. Saman Hosseinpour[#], **Fujie Tang**[#], Fenglong Wang, Ruth A. Livingstone, Simon J. Schlegel, Tatsuhiko Ohto, Mischa Bonn, Yuki Nagata, and Ellen H. G. Backus. Chemisorbed and Physisorbed Water at the TiO₂/Water Interface. *J. Phys. Chem. Lett.*, 2017, **8**, 2195. ([#]equal contribution)
32. **Fujie Tang**, Tatsuhiko Ohto, Taisuke Hasegawa, Mischa Bonn and Yuki Nagata. $\pi^+-\pi^+$ Stacking of Imidazolium Cations Enhances Molecular Layering of Room Temperature Ionic Liquids at Their Interfaces. *Phys. Chem. Chem. Phys.*, 2017, **19**, 2850.
33. Fivos Perakis, Luigi De Marco, Andrey Shalit, **Fujie Tang**, Zachary R. Kann, Thomas D. Kuhne, Renato Torre, Mischa Bonn, and Yuki Nagata. Vibrational Spectroscopy and Dynamics of Water. *Chem. Rev.*, 2016, **116**, 7590.

BOOKS & BOOK CHAPTERS

1. **Fujie Tang**. Structures and Dynamics of Interfacial Water: Input from Theoretical Vibrational Sum-frequency Spectroscopy. Springer Thesis Series 2019 (Recognizing Outstanding Ph.D. Research), ISBN 978-981-13-8964-1, by Springer Nature Singapore.
2. **Fujie Tang**, Takakazu Seki, Chun-Chieh Yu, Yuki Nagata. Microscopic Structure of Ice Surface Viewed through Sum-frequency Generation Spectroscopy. Chapter 3 of “Chemistry of the Cryosphere”, Advances in Atmospheric Chemistry, Vol. 3, pp. 139-171 (2021), by World Scientific Publishing.
3. **Fujie Tang**, Xifan Wu. Theoretical X-ray Absorption Spectroscopy of Liquid Water Using First Principles Calculations. Chapter 1 of “Properties of Water from Numerical and Experimental Perspectives”, pp. 1-24 (2022), by CRC Press.

INVITED TALKS & SEMINARS

1. **Fujie Tang**. Invited Seminar Talk. China Spallation Neutron Source, CSNS, Dongguan, Guangdong, China 2025.12
2. **Fujie Tang**. Invited Talk. The 1st National Conference on Surface and Interface Science, Chengdu, Sichuan, China 2025.05
3. **Fujie Tang**. Invited Talk. The 4th National Conference of Ultrafast Vibrational Spectroscopy, Xi'an, Shaanxi, China 2025.04

4. **Fujie Tang.** Invited Talk. The 15th National Conference of Theoretical and Computational Chemistry, Changchun, Jilin, China 2024.08
5. **Fujie Tang.** Invited Talk. The 13th National Conference on Soft Matter and Biological Physics, Xi'an, Shaanxi, China 2024.04
6. **Fujie Tang.** Invited Talk. American Physical Society, March Meeting, Minneapolis, MN, USA 2024.03
7. **Fujie Tang.** Invited Seminar Talk (virtual). Department of Physics, University of California at Berkeley, Berkeley, CA, USA 2021.07
8. **Fujie Tang.** Invited Seminar Talk. Department of Mechanical Engineering, Stanford University, Stanford, CA, USA 2019.07
9. **Fujie Tang.** Invited Seminar Talk. Department of Physics, University of California at Berkeley, Berkeley, CA, USA 2019.06
10. **Fujie Tang.** Invited Talk. The 11th National Soft Matter Physics Conference, Xiamen University, Xiamen, China 2017.03

ADVISING &
MENTORING

Current Ph.D. Advisees

1. Ms. Qi You, coadvised with Prof. Jun Cheng 2023
2. Mr. Boying Huang 2024
3. Mr. Jiezhen Xia 2024
4. Mr. Xianglong Du, coadvised with Prof. Jun Cheng 2025
5. Mr. Fanjie Xu, coadvised with Prof. Jun Cheng 2025
6. Ms. Wanlin Xu 2025

Current Graduate Advisees

1. Mr. Pengwei Xu, coadvised with Prof. Jun Cheng 2023
2. Mr. Suyang Zhong 2024
3. Ms. Lu Cao 2025
4. Mr. Yuyang Zhu 2025

JOURNAL
REVIEWER

Physical Review Letters, Physical Review Series, Journal of Physical Chemistry A/B/C/Letters, Chemical Physics Letters, Journal of Computational and Theoretical Chemistry, International Journal of Quantum Chemistry, Journal of Colloid and Interface Science

PROFESSIONAL
SOCIETY
MEMBERSHIPS

American Physics Society, American Chemical Society, Chinese Chemical Society

REFERENCES

Prof. Limei Xu

Professor

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Prof. Xifan Wu

Professor

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Prof. Mischa Bonn

Director

Max Planck Institute for Polymer Research, Mainz, Germany

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